



Research paper

Configurational coupled cluster approach with applications to magnetic model systems

Siyuan Wu, Marcel Nooijen *

Department of Chemistry, University of Waterloo, 200 University Avenue West, Waterloo, Ontario N2L 3G1, Canada

ARTICLE INFO

Article history:

Received 10 January 2018

In final form 6 April 2018

Available online 8 April 2018

Keywords:

Coupled cluster

Effective Hamiltonian

Configurational space

Magnetic model system

ABSTRACT

A general exponential, coupled cluster like, approach is discussed to extract an effective Hamiltonian in configurational space, as a sum of 1-body, 2-body up to n-body operators. The simplest two-body approach is illustrated by calculations on simple magnetic model systems. A key feature of the approach is that equations up to a certain rank do not depend on higher body cluster operators.

© 2018 Elsevier B.V. All rights reserved.

1. Introduction

The theoretical investigation of magnetic systems is an important challenge in quantum chemistry. Current approaches essentially partition the problem into two manageable pieces. Given a magnetic material, the system is decomposed in suitable interacting magnetic dimers and J-coupling constants for each dimer of interest are extracted from ab initio calculations or broken symmetry density functional theory (DFT) [1,2] calculations. The recent review by Malrieu et al. [3] provides an in-depth description of suitable methods, as does the recent monograph by Broer et al. [4]. Once a magnetic model Hamiltonian based on a decomposition in terms of dimers is extracted, the resulting model Hamiltonian can be treated using methods suitable for spin-Hamiltonian, e.g. the density matrix renormalization group (DMRG) approach [5,6].

The problem investigated in this paper is perhaps best introduced through an example. Consider, we have a number of magnetic moieties, weakly coupled through some spacers. The magnetic moieties might be individual atoms, e.g. N, O, F or Cr, while the spacers in this paper will be represented by Ar atoms. Each magnetic moiety can be represented by a small set of localized (atomic) orbitals that comprise a complete active space or CAS. The exact solution to this model (CASCI) problem is obtained by performing a full CI calculation over the CAS configurations. This quickly becomes prohibitively expensive, and our goal is to find a more economical solution. Each magnetic moiety is characterized by a number of low-lying states, that are multiconfigura-

tional in nature, while in addition there are a larger number of states for each moiety that are substantially higher in energy. The CASCI solution itself is not sufficiently accurate, and inclusion of dynamical correlation effects is vital. We will disregard this complication here but anticipate that inclusion of dynamical correlation is possible for the approach, e.g. using the recently developed multireference equation of motion (MREOM) approach [7–9].

The aim we have is to find an effective Hamiltonian over only the low-lying states that can capture the coupling between the low-lying states in a systematic fashion based on a Bloch equation [10,11]. The effective Hamiltonian may contain higher-body interactions, even if the original Hamiltonian contains only two-body interactions, i.e. only dimer interactions. At the heart of the procedure is a transformation parameterized by an exponential operator. However, the operator is not expressed in the language of second quantization, but rather using the configurations on each site, which can be partitioned in the low-lying states of interest and the remaining high-lying states. The approach is coined Configurational Coupled Cluster therefore, and the aim is to find a systematic effective Hamiltonian acting in the subspace of low-lying states. The subsequent diagonalization of this effective Hamiltonian is not trivial if a large number of magnetic sites is considered, and in future work we plan to address the direct calculation of thermal properties for such effective Hamiltonians. This paper is primarily a first investigation of the ideas, and it is organized as follows. In Section 2, the theory is discussed, and we start by an introduction of the magnetic model systems in this work, that provides motivation. We continue to discuss the general theory and its salient features. We explicitly consider the simplest two

* Corresponding author.

E-mail address: nooijen@uwaterloo.ca (M. Nooijen).

realizations of the theory using a pair $\hat{T}_2$ operator and the extraction of two-body and three-body effective Hamiltonians. In the results section, we evaluate and illustrate the methodology for artificial, but sufficiently complicated model systems that have up to four magnetic sites. The final section summarizes the results and provides further perspectives.

2. Theory

2.1. Introduction to magnetic model systems

Let us first introduce some artificial magnetic systems, such as Ar^2N_3 , Ar_2O_3 , Ar_2Cr_3 , that cannot be made experimentally, but they serve as illustrations and useful model systems. These systems are described as open-shell systems, as nitrogen, oxygen and chromium atoms all have unpaired electrons. In addition, the spin of unpaired electrons can create a magnetic field; as a result, each N, O, Cr, atom is considered as a magnetic site. Meanwhile, Ar atoms act as spacers. The geometry of the system is designed such that the magnetic atoms are well separated.

For each magnetic system, we would like to preserve a fixed number of particles on each magnetic site, or equivalently we will neglect ionic configurations. At the level of the second-quantized Hamiltonian, the number of creation and annihilation operators for each site should be equal. We assume that

$$h_p^r = 0, \quad \text{unless } (p, r) \in (i) \quad (1)$$

$$h_{pq}^{rs} = 0, \quad \text{unless } (p, q, r, s) \in (i) \text{ or } \begin{cases} (p, r) \in (i), (q, s) \in (j) \\ (p, s) \in (i), (q, r) \in (j) \end{cases} \quad (2)$$

Here we use indices i, j to label two different magnetic sites. Localized orbitals centered on magnetic sites are labelled as p, q, r, s . As a result, neutral configurations cannot couple to ionic configurations. This form of the Hamiltonian is an important simplifying assumption. For more realistic systems, one might employ a preliminary similarity transformation such that ionic terms are transformed to zero, in the spirit of the MREOM approach [7–9]. Alternatively, the current approach would have to be extended to include ionic configurations. Both solutions are non-trivial, and at this point we will simply analyze results under the simplifying assumption that Eqs. (1) and (2) are valid. Another significant simplification arises if we assume that we can limit the orbitals to a relatively small set of active orbitals that describe the primary magnetic interactions. In practice, one can solve complete active space self-consistent field (CASSCF) [12] solutions for a small number of high-spin multiplets, that do not require a large diagonalization space. We will consider here the problem of calculating *all* spin states in the CAS space, while neglecting ionic interactions, using a basis of localization orbitals. This provides a working hypothesis for our initial investigations of the Configurational Coupled Cluster approach. Finally, spin-orbit coupling is important in these systems [13], but is not considered in this preliminary study.

The new neutral Hamiltonian in a complete active space (CAS) is given by

$$\hat{H} = \sum_i \sum_{p,p'} h_{p,p'}^p \hat{p}^\dagger \hat{p}' + \frac{1}{2} \sum_i \sum_{p_1,p_2,p_3,p_4} h_{p_1,p_2,p_3,p_4}^{p_1 p_2} \hat{p}_1^\dagger \hat{p}_2^\dagger \hat{p}_3 \hat{p}_4 + \frac{1}{2} \sum_{ij} \sum_{p,q,p',q'} h_{p,q,p',q'}^{pq} \hat{p}^\dagger \hat{q}^\dagger \hat{q}' \hat{p}' + \frac{1}{2} \sum_{ij} \sum_{p,q,p',q'} h_{p,q,p',q'}^{qp} \hat{q}^\dagger \hat{p}^\dagger \hat{q}' \hat{p}' \quad (3)$$

The labels $p, p', p_1, p_2, p_3, p_4$ denote localized orbitals on site i , while q, q' indicate localized orbitals on site j . The fourth term in Eq. (3) is the exchange term arising from the two-body integral, which is mainly responsible for magnetic interactions. The spin can be exchanged between two sites through action of the two-body operators (see Fig. 1).

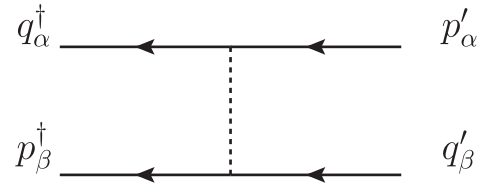


Fig. 1. Diagram representing the exchange interaction.

In order to investigate the magnetism of these model systems, the properties with regard to the electronic configuration of each atom should be described first. We define many-electron low-lying states corresponding to different atoms (or sites) $i, j, k, l, \dots$ as $|I\rangle, |J\rangle, |K\rangle, |L\rangle, \dots$, and high-lying states are denoted as $|A\rangle, |B\rangle, |C\rangle, |D\rangle, \dots$ while the general states are labelled as $|P\rangle, |Q\rangle, |R\rangle, |S\rangle, \dots$, respectively. Likewise, if there is more than one state, we can use the general labels $|I'\rangle, |J'\rangle, |K'\rangle, |L'\rangle, \dots$, for the low-lying states, $|A'\rangle, |B'\rangle, |C'\rangle, |D'\rangle, \dots$, standing for high-lying states, and $|P'\rangle, |Q'\rangle, |R'\rangle, |S'\rangle, \dots$, describing general states for each different site. Importantly, this partitioning of configurations is based on single site (or atomic) calculations, using the localized orbitals obtained after the high-spin CASSCF calculation (see Table 1).

To continue our discussion of the magnetic model systems, we first consider the electronic configuration for some representative atoms in Table 2. We can quantitatively explore the number of microstates of the magnetic model systems Ar_2X_3 , $\text{X} = \text{N}, \text{O}, \text{Cr}$, listed in Table 3, which indicates the complexity of the problem, even for simple model systems.

Our aim is to reduce the size of the full space of CASSCF dimension, to that of the low-lying states. For example, the number of valence states is 48,620 for Ar^2N_3 , while the number of low-lying states is only 64, which would be the dimension of the two-body effective Hamiltonian.

Table 1

Electronic configuration representation for systems up to four sites.

Sites	Low-lying States	High-lying States	General States
i	$ I\rangle, I'\rangle, I''\rangle, \dots$	$ A\rangle, A'\rangle, A''\rangle, \dots$	$ P\rangle, P'\rangle, P''\rangle, \dots$
j	$ J\rangle, J'\rangle, J''\rangle, \dots$	$ B\rangle, B'\rangle, B''\rangle, \dots$	$ Q\rangle, Q'\rangle, Q''\rangle, \dots$
k	$ K\rangle, K'\rangle, K''\rangle, \dots$	$ C\rangle, C'\rangle, C''\rangle, \dots$	$ R\rangle, R'\rangle, R''\rangle, \dots$
l	$ L\rangle, L'\rangle, L''\rangle, \dots$	$ D\rangle, D'\rangle, D''\rangle, \dots$	$ S\rangle, S'\rangle, S''\rangle, \dots$

Table 2

Electronic configuration for three different atoms.

Atom-Type	Ground States Symbol	Degeneracy	Number of Neutral States Per Atom
N	$4s$	4	$\binom{6}{3} = 20$
O	$3p$	9	$\binom{6}{4} = 15$
Cr	$7s$	7	$\binom{12}{6} = 924$

Table 3

Microstates for Ar_2X_3 , $\text{X} = \text{N}, \text{O}, \text{Cr}$.

Molecule	Total Valence States	Total Neutral States	Total Low-lying States
Ar^2N_3	$\binom{18}{9} = 48620$	$20^3 = 8000$	$4^3 = 64$
Ar_2O_3	$\binom{18}{12} = 18564$	$15^3 = 3375$	$9^3 = 729$
Ar_2Cr_3	$\binom{36}{18} = 9.07\text{E}+09$	$924^3 = 7.89\text{E}+08$	$7^3 = 343$

2.2. General formulation of configurational coupled cluster approach

We assume that the neutral molecular magnetic Hamiltonian in configuration space is expressed as

$$\hat{H} = \sum_i \sum_{P,P' \in i} H_p^{P'} |P'\rangle \langle P| + \sum_{i < j} \sum_{P,P' \in i} \sum_{Q,Q' \in j} H_{pq}^{P'Q'} |P'Q'\rangle \langle PQ| \quad (4)$$

We can define the *effective Hamiltonian* $\hat{G}$ over the low-lying states by

$$\hat{G} = \hat{G}_1 + \hat{G}_2 + \hat{G}_3 + \dots \quad (5)$$

where each corresponding term in $\hat{G}$ is given by

$$\hat{G}_1 = \sum_i \sum_{I,I' \in i} G_i^{I'} |I'\rangle \langle I| \quad (6)$$

$$\hat{G}_2 = \sum_{i < j} \sum_{I,I' \in ij} \sum_{J,J' \in j} G_{ij}^{I'J'} |I'J'\rangle \langle IJ| \quad (7)$$

$$\hat{G}_3 = \sum_{i < j < k} \sum_{I,I' \in ij} \sum_{J,J' \in jk} \sum_{K,K' \in k} G_{ijk}^{I'J'K'} |I'J'K'\rangle \langle IJK| \quad (8)$$

To obtain equations for the effective Hamiltonian $\hat{G}$, we will employ a Bloch equation. To this end, we introduce a projection operator on the model space of low-lying states.

$$\hat{P} = \sum_{I,J,K} |IJK \dots\rangle \langle IJK \dots| \quad (9)$$

The operator $\hat{Q} = \hat{1} - \hat{P}$, is a projector on states in which at least one low-lying state is replaced by a high-lying state. The operator $\hat{W}$ is the wave operator entering the Bloch equation in intermediate normalization, hence $\hat{P}\hat{W}\hat{P} = \hat{1}$. The Bloch equation reads

$$\hat{H}\hat{W}\hat{P} - \hat{W}\hat{P}\hat{G}\hat{P} = 0 \quad (10)$$

hence

$$\hat{P}\hat{H}\hat{W}\hat{P} = \hat{P}\hat{W}\hat{P}\hat{G}\hat{P} = \hat{P}\hat{G}\hat{P} = \hat{G} \quad (11)$$

and

$$\hat{Q}\hat{H}\hat{W}\hat{P} - \hat{Q}\hat{W}\hat{G}\hat{P} = 0 \quad (12)$$

If the wave operator satisfies the Bloch equation, eigenvalues of $\hat{G}$ are eigenvalues of the full $\hat{H}$, and if $|\lambda\rangle$ is an eigenstate of $\hat{G}$ in the model space ($\hat{G}|\lambda\rangle = |\lambda\rangle E_\lambda$), then

$$\hat{H}(\hat{W}|\lambda\rangle) = \hat{W}\hat{G}|\lambda\rangle = \hat{W}|\lambda\rangle E_\lambda \quad (13)$$

Importantly, our goal is to have access to *all* low-lying states, and the use of Bloch equation is essential. The next step is to introduce an exponential-like parameterization for $\hat{W}$, using a cluster expansion. The cluster operator in configuration space is defined as

$$\hat{T} = \hat{T}_1 + \hat{T}_2 + \hat{T}_3 + \dots \quad (14)$$

where each contribution of $\hat{T}$ is expressed as

$$\hat{T}_1 = \sum_i \sum_{I,A \in i} T_i^A |A\rangle \langle I| \quad (15)$$

$$\hat{T}_2 = \sum_{i < j} \sum_{I,P \in ij} \sum_{Q \in j} T_{ij}^{PQ} |PQ\rangle \langle IJ| \quad (16)$$

$$\hat{T}_3 = \sum_{i < j < k} \sum_{I,P \in ij} \sum_{Q \in jk} \sum_{R \in k} T_{ijk}^{PQR} |PQR\rangle \langle IJK| \quad (17)$$

The general labels P, Q and P, Q, R in Eqs. (16) and (17) exclude terms in which all labels refer to low-lying states, i.e. we assume the corresponding cluster amplitudes are zero. The $\hat{T}$ -operators in

general do not all commute due to the presence of mixed excitation operators, e.g. $\hat{T}_{ij}^{B}$. For this reason, we define the wave operator in the Bloch equation as

$$\hat{W} = \{e^{\hat{T}}\} \quad (18)$$

Here a “normal-ordered” exponential operator is defined such that terms in which operators act on a given site more than once are explicitly excluded, or

$$\{e^{\hat{T}}\} = \exp(\hat{T}') \quad (19)$$

where the prime indicates removal of the undesired terms. Hence, for example, $\hat{T}_{jk}^{BC} \hat{T}_{ij}^{AJ'}$ is excluded, because the excitation operates on site j twice.

Substituting in the Bloch equation, we get

$$\hat{Q}\hat{H}\{e^{\hat{T}}\}\hat{P} - \hat{Q}\{e^{\hat{T}}\}\hat{G}\hat{P} = 0 \quad (20)$$

or

$$\hat{Q}\left\{\left(\hat{H}\{e^{\hat{T}}\}\right)_{\text{connected}} e^{\hat{T}}\right\} - \left\{\left(\{e^{\hat{T}}\}\hat{G}\right)_{\text{connected}} e^{\hat{T}}\right\}\hat{P} = 0 \quad (21)$$

To derive Eq. (21) from Eq. (20), we note that $\hat{H}$ can have sites in common with zero, one or two $\hat{T}$ -operators, and this is indicated by $(\hat{H}\{e^{\hat{T}}\})_{\text{connected}}$, the remaining $\hat{T}$ -operators would not have sites in common with $\hat{H}$. The term $\{e^{\hat{T}}\}\hat{G}$ can be analyzed in a similar fashion. For convenience, we can identify a connected residual operator

$$\hat{R} = \left(\hat{H}\{e^{\hat{T}}\}\right)_{\text{connected}} - \left(\{e^{\hat{T}}\}\hat{G}\right)_{\text{connected}} \quad (22)$$

$$\hat{R} = \hat{R}_1 + \hat{R}_2 + \hat{R}_3 + \dots \quad (23)$$

where each contribution of $\hat{R}$ is expressed as

$$\hat{R}_1 = \sum_i \sum_{I,A \in i} R_i^A |A\rangle \langle I| \quad (24)$$

$$\hat{R}_2 = \sum_{i < j} \sum_{I,P \in ij} \sum_{Q \in j} R_{ij}^{PQ} |PQ\rangle \langle IJ| \quad (25)$$

$$\hat{R}_3 = \sum_{i < j < k} \sum_{I,P \in ij} \sum_{Q \in jk} \sum_{R \in k} R_{ijk}^{PQR} |PQR\rangle \langle IJK| \quad (26)$$

In terms of the connected residual operator $\hat{R}$, the Bloch equation can be written as

$$\hat{Q}\{\hat{R}e^{\hat{T}}\}\hat{P} = 0 \quad (27)$$

Clearly, the Bloch equation is satisfied if

$$\hat{Q}\hat{R}\hat{P} = 0 \quad (28)$$

and for each individual connected component in $\hat{R}$, we have

$$R_i^A = 0 \quad (29)$$

$$R_{ij}^{PQ} = 0 \quad (30)$$

$$R_{ijk}^{PQR} = 0 \quad (31)$$

As before, we exclude terms in which all labels P, Q, R refer to low-lying states. These equations have an important hierarchical feature. $\hat{R}_n$ only depends on $\hat{T}_m$ or $\hat{G}_m$ -elements that are at most n -body. Hence, one can solve the equation in sequence. One first solves T_1 and G_1 amplitudes from single-site calculations; and then obtains T_2 and G_2 amplitudes from two-site calculations using the resulting T_1 and G_1 amplitudes. Importantly, higher rank amplitudes do not contribute to lower rank equations. All the terms in the residual equation have the same overall rank, or refer to a

definite n -body equation. This is a simplifying feature that is very different from the usual Coupled Cluster approach in second quantization [14].

In a recent paper, similar ideas have been presented as Excitonic Coupled Cluster theory [15,16]. The theory underlying Excitonic Coupled Cluster is general and a major focus is the general representation of the Hamiltonian in excitonic or configurational form. In this work, we essentially massage and approximate the Hamiltonian to have the simple form of Eq. (4). A major difference compared to Excitonic Coupled Cluster approach is that we are interested in the energies of the full manifold of low-lying states, and this requires a multi-state theory based on Bloch equation. Excitonic Coupled Cluster theory is currently a ground state theory akin to conventional single reference Coupled Cluster theory [17].

While the equations can in principle be solved in the above fashion, the procedure might lead to large T_1 and T_2 amplitudes, and this is undesirable, as it would mean 3- and higher-body contribution can be expected to be large. In the modified procedure below, we take advantage of the fact that the partitioning into 0-, 1-, 2- and higher-body operators is to some extent ambiguous. For example, in conventional Coupled Cluster approaches, one first solves a mean-field problem and writes the Hamiltonian in terms of the one-body Fock operator and a two-body fluctuation potential. An analogous approach is pursued here. We first solve configurational mean-field equations, and partition the Configurational Hamiltonian in a zero-order mean-field energy, a one-body mean-field operator and a 2-body fluctuation term. The partitioning of states into low-lying and high-lying states is also determined from the mean-field calculation. All matrix-elements are transformed to the mean-field configurational basis, while setting $T_1 = 0$. This mean-field calculation step replaces the first step in the general procedure. In the second step, we solve for the T_2 -amplitudes capitalizing on the fact that the approach is exact for two-body systems, while the exact T_2 equations do not depend on higher-order amplitudes. Hence rather than solving for non-linear quadratic equations, we extract the T_2 -amplitudes from an exact diagonalization of the two-body part of the Hamiltonian as expressed in a mean-field form. Both procedures are discussed in more detail below.

2.3. Step 1: mean-field calculation and conversion of Hamiltonian to mean-field form

Assume that the Hamiltonian with respect to configuration space is written as

$$\hat{H} = \hat{H}_1 + \hat{H}_2 = \sum_i \sum_{P,P' \in i} H_P^{P'} |P'\rangle \langle P| + \sum_{i < j} \sum_{P,P' \in i} \sum_{Q,Q' \in j} H_{PQ}^{P'Q'} |P'Q'\rangle \langle PQ| \quad (32)$$

We will try to find low-lying single-site states that are linear combinations of basis states

$$|I\rangle = \sum_P |P\rangle C_{PI} \quad (33)$$

This allows us to define a normalized density matrix

$$\hat{D}_P^{P'} = \frac{1}{N_I} \sum_{I=1}^{N_I} C_{PI} C_{P'I} \quad (34)$$

in terms of orthonormal states

$$\sum_P C_{PI} C_{P'I'} = \delta_{II'} \quad (35)$$

Therefore, we obtain the trace of the density matrix

$$\text{Tr}(\hat{D}) = 1 \quad (36)$$

The density matrix can be used to define a mean-field one-site Hamiltonian or Fock operator.

$$F_P^{P'}(i) = H_P^{P'}(i) + \sum_j \sum_{Q,Q'} H_{PQ}^{P'Q'} D_Q^{Q'} \quad (37)$$

The orthonormal coefficients C_{PI} are obtained by diagonalizing the Fock operator. Hence we can define a self-consistent field procedure:

$$FC = C\epsilon \quad (38)$$

$$D = \frac{1}{N_I} \sum_I CC^T \quad (\text{see Eq. (34)}) \quad (39)$$

$$F = H_1 + H_2 D \quad (\text{see Eq. (37)}) \quad (40)$$

This procedure in effect minimizes the mean-field energy

$$E_{SCF} = \sum_i \sum_{P,P' \in i} H_P^{P'} D_P^{P'} + \sum_{ij} \sum_{P,P' \in i} \sum_{Q,Q' \in j} H_{PQ}^{P'Q'} D_P^{P'} D_Q^{Q'} \quad (41)$$

This procedure converges for all the systems we present in the results section, and it is analogous to a Hartree Fock mean-field calculation in electronic structure theory. This calculation is the first step in the procedure and is used to obtain the precise partitioning between low-lying and high-lying states for a single site. In Appendix A, we discuss below the Hamiltonian can be rewritten in a mean-field form.

$$\hat{H} = E_{SCF} + \sum_i \tilde{f}_i + \sum_{i < j} \tilde{V}_{ij} \quad (42)$$

where the so-called normal ordered operators are defined such that partial traces over the mean-field density matrices vanish.

It is not clear at present if the mean-field plus normal ordering of the Hamiltonian is vital for the procedure to work well. In anticipated more general applications of Configurational Coupled Cluster, we think it probably will be a vital step to ensure that two-body transformation amplitudes are small. We have verified explicitly that diagonalization of the redefined mean-field Hamiltonian yields the same excitation energies as the original Hamiltonian. Hence we verified for up to four-body systems that the procedure is valid in principle.

2.4. Step 2: solution of T_2 amplitudes using exact pairwise diagonalization approach

For each pair of magnetic sites, we construct the pair Hamiltonian matrix

$$M_{PQ}^{P'Q'} = \tilde{f}_P^{P'} \delta_Q^{Q'} + \delta_P^{P'} \tilde{f}_Q^{Q'} + \tilde{V}_{PQ}^{P'Q'} \quad (43)$$

using the normal ordered form derived in Appendix A. The two-body T -amplitudes are required to exactly block diagonalize the two-body Hamiltonian, and in general such a block transformation can easily be extracted using exact diagonalization, see for example the discussion in [18]. We partition the Hamiltonian in a block form using labels M and O , where M indicates model space components, while O represents the orthogonal complements. The eigenstates dominated by configurations M in the model space can be written as $\begin{pmatrix} C_{M\lambda} \\ C_{O\lambda} \end{pmatrix}$. One can solve

$$\begin{pmatrix} \mathbb{1} \\ T \end{pmatrix} (C_{M\lambda}) = \begin{pmatrix} C_{M\lambda} \\ C_{O\lambda} \end{pmatrix} \quad (44)$$

or $T_{OM} = C_{O\lambda} (C_{M\lambda})^{-1}$. It is straightforward to verify that $\hat{W} = (\hat{\mathbb{1}}_{MO} + \hat{T}_{OM})$ satisfies the matrix analog of Bloch equation. This

is the essence of the procedure. If we apply this strategy to the two-body problem at hand we get, in more detail: diagonalize the M matrix and determine $N_I * N_J$ low-lying eigenvectors C_{PQ}^λ , where $\lambda = 1, \dots, N_I * N_J$. We first collect the low-lying square matrix $\tilde{C}_{IJ}^\lambda = C_{IJ}^\lambda$, and then invert this matrix to obtain S_λ^{IJ} . Then we can define the $\tilde{T}$ -amplitudes as

$$\tilde{T}_{PQ}^{IJ} = \sum_\lambda C_{PQ}^\lambda S_\lambda^{IJ} \quad (45)$$

It can be easily verified that

$$C_{PQ}^\lambda = \tilde{T}_{PQ}^{IJ} C_{IJ}^\lambda = \sum_{\mu, I, J} C_{PQ}^\mu (C_{IJ}^\mu)^{-1} C_{IJ}^\lambda \quad (46)$$

As $\tilde{T}_{IJ'}^{IJ} = \delta_I^I \delta_{J'}^J$, we finally define excitation operator $\hat{T} = \tilde{T}$, except that we explicitly set all $T_{IJ'}^{IJ} = 0$. If we only consider the two-site Hamiltonian, we obtain

$$\hat{G} = (M\tilde{T})_{low-lying} = [\hat{M}(\hat{1} + \hat{T})]_{low-lying} \quad (47)$$

and it is easily verified that diagonalizing $\hat{G}$ over the low-lying states reproduces the exact low-lying eigenvalues of the full pair Hamiltonian. This method is denoted as independent pair definition of $\hat{T}$, since it is obtained by a sequence of independent pair calculations. We can obtain the pairwise G -elements in this first approximation, but the full $\hat{G}$ also contains 3- and 4-body operators. The mathematical expression for 3- and 4-body contribution to $\hat{G}$ can be given by

$$\hat{G}_{IJK}^{I'J'K'} = \sum_B \tilde{V}_{BK}^{I'J'K'} \hat{T}_{IJ}^{IB} + \text{permutations} \quad (48)$$

$$\hat{G}_{IJKL}^{I'J'K'L'} = \sum_{CD} \tilde{V}_{CD}^{I'J'K'L'} \hat{T}_{IL}^{ID} \hat{T}_{JK}^{JC} + \text{permutations} \quad (49)$$

3. Computational details

We will explore a number of variations of the methodology, mainly for the purpose of analyzing the importance of various effects. The model systems are chosen such that one can identify a number of low-lying states and there is a fair gap to the next higher states. Here we provide the first several steps in the calculation.

Step 1. Perform electronic structure calculation (CASSCF/CASCI) using the 6-31G* basis set [19,20] to extract the magnetic Hamiltonian parameters (see Appendix A).

Step 2. Identify the number of low-lying and high-lying states for each magnetic site, and perform mean-field calculation.

Step 3. Solve the independent pair T_2 amplitudes by diagonalizing the pair Hamiltonians.

These steps are always the same, next we can introduce some different approximations.

In the first, zeroth order or bare H approach, we set T_2 amplitudes equal to zero, and hence simply diagonalize $\hat{H}$ over the low-lying configurations. In the second approach, that we label “Heff_2-body”, the $\hat{G}_2$ operator is calculated for each pair, but no 3- and 4-body effects are considered. Diagonalizing the $\hat{G}_2$ operator for a given pair exactly reproduces the pair energies of the corresponding full Hamiltonian. In the third approximation, that we call “Heff_2/3-body”, the $\hat{G}_3$ operator is included for magnetic systems that have three and four magnetic sites. This $\hat{G}_3$ operator is defined using the pair $\hat{T}_2$ operator, but does not depend on $\hat{T}_3$. This will give us an idea of the importance of higher-body effects.

4. Results

In this section, we test the methodology for magnetic model systems that consist of an Ar atom that acts like a spacer, and this is surrounded by open-shell magnetic atoms, O, N, F and H. Oxygen atom has 9 low-lying states (3P ground state) and 15 states in total. Likewise, N atom has 4 low-lying states (4S) and 20 states in total, while for F and H atoms, we would simply have 6 low-lying states and 2 low-lying states, respectively. It turns out that the effective Hamiltonian approach is highly accurate if there is a large energy gap. The ground state energy is set equal to zero and we focus on excitation energies. For example, for NArNN magnetic system, the energy gap between low-lying and high-lying states is around 2.7 eV or 22,000 cm^{-1} .¹ In Table 4, we report the low-lying states energies for NArNN using the various approximations, bare H, Heff_2-body, Heff_2/3-body and compared to the exact energies. This system has spin symmetry, and therefore we obtain a multiplet structure. In the table, we present the energy for each magnetic level and the degeneracy of each energy level. For the Heff_2/3-body approach, the degeneracy pattern is slightly broken, therefore we report the *average energy* in each multiplet, and the energy spread is defined as $E_{\text{highest}} - E_{\text{lowest}}$ within a multiplet. From Table 5, it can be seen that the statistical errors for each variant of effective Hamiltonian approach are extremely small. This indicates that decoupling all low-lying magnetic states from the other configurations is not a challenging benchmark for systems that have a large energy gap. This is an important result as the magnetic systems in which we are interested have this feature that the low lying magnetic states are well separated from the high lying states. However, this does not seem to be very challenging, as even diagonalizing the bare Hamiltonian gives good results. As a result, we need to find more challenging systems by identifying a gap within the magnetic manifold of states and focusing the effective Hamiltonian approach on the low-lying states within the magnetic manifold. The nitrogen magnetic site is not useful as all p-orbitals in N are occupied with a single electron. However, the population of the p-orbitals in the F and O atoms is uneven and this provides interesting possibilities. The oxygen and fluorine atoms are significantly perturbed by the Ar atom. This means that the p-orbitals are not equivalent. For example, at a bond distance (Ar-F) of 2.9 Å, the spectrum for ArF is split into 2 low-lying states and 4-high-lying states with a gap of around 174 cm^{-1} .² Likewise, the magnetic states in ArO can be subdivided into 6 low-lying states and 3 high-lying states with a gap of about 386 cm^{-1} .³ The number of high-lying and low-lying states for each atom is indicated in Table 6. In the next step, we will consider magnetic model systems for the more challenging test. Table 7 indicates the number of states and energy gap of each system.⁴

4.1. Magnetic dimers: FArF and FArO

As can be seen in Tables 8 and 9, the Heff_2-body approach reproduces the exact low-lying states energies for FArO and FArF. This is anticipated, as there are no 3-body and 4-body effects for magnetic dimers. This can be viewed as simply a check of the code.

4.2. Magnetic trimers and tetramers: FArFF, FArOF, HArFFF and HArOFF

In this section, we will consider a number of magnetic model systems that contain up to 4 sites: FArFF, FArOF, HArOFF and

¹ The cartesian coordinate of this molecule is provided in Supplementary material.

² Calculated by CASCI using the basis set of 6-31G* in ACES II package.

³ Calculated by CASCI using the basis set of 6-31G* in ACES II package.

⁴ The cartesian coordinate of each system is given in Supplementary Material.

Table 4Comparison of low-lying states energies for different approaches of NArNN. All results are in cm^{-1} .

Level	Degeneracy	Exact	Bare H	Heff_2-body	Heff_2/3-body	
					Avg-Energy	Spread
1	10	0.000	0.000	0.000	0.000	0.000
2	16	60.133	60.213	60.127	60.134	0.002
3	12	106.935	107.046	106.933	106.935	0.008
4	6	106.992	107.046	106.986	106.992	0.000
5	4	140.339	140.498	140.349	140.337	0.000
6	4	140.415	140.498	140.419	140.413	0.000
7	8	140.453	140.498	140.454	140.452	0.000
8	4	160.409	160.569	160.426	160.405	0.002

Table 5Statistical analysis of effective Hamiltonian data compared to exact low-lying states energies for NArNN. All results are in cm^{-1} .

Molecule	Statistical Errors	Bare H	Heff_2-body	Heff_2/3-body
NArNN	AVG	0.086	0.006	0.001
	MAX	0.160	0.017	0.004
	RMS	0.101	0.008	0.001

Table 6

The number of low-lying and high-lying magnetic states for O, F and H atom with the inclusion of Argon atom.

Atom	Number of low-lying magnetic states	Number of high-lying magnetic states
O	6	3
F	2	4
H	2	0

HArFFF. In Tables 10–13, we report the degeneracies and excitation energies of low-lying states for each magnetic system.

Table 14 shows the statistical errors of these approaches compared to exact low-lying states excitation energies for magnetic tri-

mers and tetramers. If we examine the %RMS error in Table 14, we see significant improvements between the bare H and Heff_2-body results. The additional improvement from including the 3-body contribution deriving from the pair $\hat{T}_2$ operators is not systematic and often insignificant. In only one case HArOFF, we do see a clear improvement. For other systems, results are insignificantly affected by 3-body contribution. It appears that to achieve a consistently improved results the effects of the $\hat{T}_3$ operator is to be considered. It may be that a perturbative estimate of $\hat{T}_3$ would be sufficiently accurate, but this is for a future investigation.

To illustrate the results of the calculation in a graphical fashion, we draw a curve of excitation energy versus state number referred

Table 7

The number of states and energy gap of magnetic model systems.

Molecules	Number of States		Energy Gap (cm^{-1})	
	Total	Low-lying	Highest low-lying state	lowest high-lying state
FArF	36	4	22.891	62.281
FArO	90	12	166.387	332.532
FArFF	216	8	13.988	99.358
FArOF	540	24	11.633	79.026
HArOFF	1080	48	38.454	85.244
HArFFF	512	16	47.894	99.437

Table 8Comparison of low-lying states energies for different approaches of FArF. All results are in cm^{-1} .

Level	Degeneracy	Exact	Bare H	Heff_2-body
1	3	0.000	0.000	0.000
2	1	22.891	26.536	22.891

Table 9Comparison of low-lying states energies for different approaches of FArO. All results are in cm^{-1} .

Level	Degeneracy	Exact	Bare H	Heff_2-body
1	4	0.000	0.000	0.000
2	2	43.930	50.309	43.930
3	4	136.334	150.889	136.334
4	2	166.386	177.422	166.386

Table 10
Comparison of low-lying states energies for different approaches of FArFF. All results are in cm⁻¹.

Level	Degeneracy	Exact	Bare H	Heff_2-body	Heff_2/3-body	
					Avg-Energy	Spread
1	4	0.000	0.000	0.000	0.000	0.000
2	4	13.988	15.809	14.032	14.033	0.000

Table 11
Comparison of low-lying states energies for different approaches of FArOF. All results are in cm⁻¹.

Level	Degeneracy	Exact	Bare H	Heff_2-body	Heff_2/3-body	
					Avg-Energy	Spread
1	5	0.000	0.000	0.000	0.000	0.000
2	3	1.292	1.352	1.294	1.293	0.000
3	1	1.936	2.032	1.943	1.941	0.000
4	3	2.859	3.411	3.127	2.648	0.000
5	5	8.313	9.615	8.586	8.534	0.000
6	3	9.451	10.789	9.750	9.701	0.000
7	1	10.018	11.371	10.328	10.281	0.000
8	3	11.633	12.940	11.649	11.907	0.000

Table 12
Comparison of low-lying states energies for different approaches of HArFFF. All results are in cm⁻¹.

Level	Degeneracy	Exact	Bare H	Heff_2-body	Heff_2/3-body	
					Avg-Energy	Spread
1	5	0.000	0.000	0.000	0.000	0.000
2	6	25.476	27.744	25.797	25.854	0.008
3	3	44.322	45.088	45.023	45.196	0.000
4	2	47.894	50.288	48.309	48.452	0.008

Table 13
Comparison of low-lying states energies for different approaches of HArOFF. All results are in cm⁻¹.

Level	Degeneracy	Exact	Bare H	Heff_2-body	Heff_2/3-body	
					Avg-Energy	Spread
1	6	0.000	0.000	0.000	0.000	0.000
2	6	5.550	6.773	5.656	5.398	0.000
3	4	8.551	8.739	8.623	8.523	0.000
4	4	9.057	9.403	9.305	8.896	0.000
5	4	13.853	15.213	14.033	13.789	0.000
6	4	14.864	16.114	14.918	14.709	0.000
7	2	15.974	16.290	16.102	15.938	0.000
8	2	21.025	22.451	21.253	21.032	0.000
9	2	26.780	27.205	27.102	26.686	0.000
10	2	28.667	29.233	29.094	28.416	0.000
11	2	31.946	33.302	32.095	31.984	0.000
12	4	33.020	33.511	33.496	32.787	0.000
13	2	34.144	33.542	34.339	34.089	0.000
14	4	38.453	39.802	39.651	38.385	0.000

Table 14
Statistical analysis of effective Hamiltonian data compare to exact low-lying states energies for magnetic model systems in this work. All results are in cm⁻¹

Molecule	Statistical Errors	Bare H	Heff_2-body	Heff_2/3-body
FArFF	AVG	0.951	0.022	0.023
	MAX	1.902	0.044	0.045
	RMS	1.345	0.031	0.032
HArFFF	AVG	1.356	0.362	0.455
	MAX	2.395	0.701	0.874
	RMS	1.696	0.440	0.554
HArOFF	AVG	0.778	0.264	0.095
	MAX	1.426	1.198	0.251
	RMS	0.924	0.394	0.123
FArOF	AVG	0.751	0.147	0.153
	MAX	1.353	0.310	0.274
	RMS	0.958	0.204	0.194

to as an excitation energy plot, and we also provide the heat capacity as a function of temperature. The heat capacity is obtained from a sum over states expression, that includes all low-lying magnetic states. In terms of each energy level E_n and temperature T [13], we define

$$Z = \sum_{n=0} e^{\frac{-(E_n - E_0)}{k_B T}} \tag{50}$$

$$P_n = \frac{1}{Z} e^{\frac{-(E_n - E_0)}{k_B T}} \tag{51}$$

$$U = \sum_n P_n E_n \tag{52}$$

$$C_\nu = \frac{1}{k_B T^2} \sum_n P_n (E_n - U)^2 \tag{53}$$

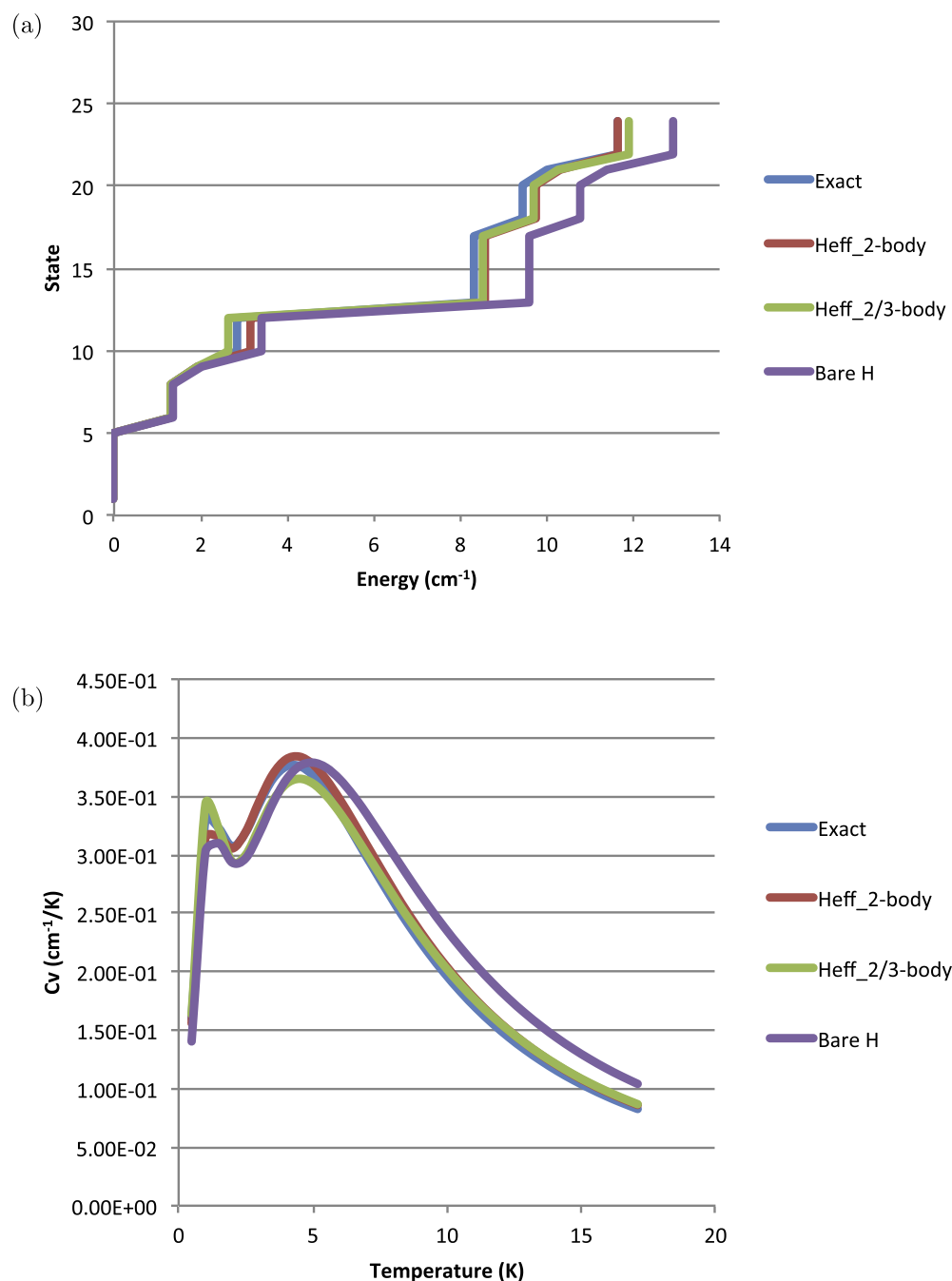


Fig. 2. Plots of (a) excitation energies and (b) heat capacity for FArOF.

In Fig. 2, we present a plot of excitation energies and heat capacity plot for the most interesting case of FArOF. It can be seen that the major improvement to the results arises from the inclusion of $\hat{T}_2$ operator. The change upon inclusion of the 3-body part of Heff on thermal properties is minor.

5. Concluding remarks

The Configurational Coupled Cluster method provides a path to extract an effective Hamiltonian over low-lying many-electron states, that are expressed as a product of localized states. The approach was applied to prototype magnetic systems, but it can be applied to in principle to other problems that have a similar

structure. For example, there is significant interest in treating rotating molecules on a lattice (e.g. molecules contained in C_{60} cages arranged on a lattice [21–23]). The theory is expected to apply to such problems too. The simplest version of the theory starts from a mean-field treatment to partition states in low-lying and high-lying states, and then uses a pair- T_2 operator to extract the effective Hamiltonian. Using only the resulting two-body effective Hamiltonian yields quite satisfactory results, for the model systems considered. However, inclusion of the implied three-body effective Hamiltonian does not systematically improve results. It would be seen preferable to include a three-body cluster operator, and use this to extract the full three-body effective Hamiltonian. An interesting aspect of the Configurational Coupled Cluster approach is that the equation for cluster amplitudes up to

rank n do not couple to equations of higher rank. Hence one can solve for T_1 , then solve for T_2 while keeping T_1 fixed, finally solve for T_3 while keeping the lower rank amplitudes fixed. The higher rank equations for T -amplitudes do depend on lower-rank amplitudes, but not conversely. This is considered as an interesting feature of the approach which is not shared by conventional Coupled Cluster approaches. In the end, the Configurational Coupled Cluster approach does not address the exponential scaling of the problem, as only the number of states per site is reduced when obtaining the effective Hamiltonian. The total number of low-lying states can grow very rapidly in magnetic systems. Therefore, in the end one is not interested primarily in the spectrum of the Hamiltonian but in thermal properties. These thermal properties may be calculated directly, and in future work we plan to address this problem using an analogous thermal Configurational Coupled Cluster approach.

Acknowledgement

This research was supported by the National Science and Engineering Research Council (NSERC) of Canada through a discovery grant to M.N.

Appendix A

A.1. Construction of configurational magnetic Hamiltonian

To calculate the matrix element of the magnetic Hamiltonian over neutral configurations, special procedures have been incorporated in the ACES II program. [24] In the first stage of the calculations, a state-averaged CASSCF/CASCI calculation is done such that all relevant configurations of the atom are included. Upon convergence of the CASSCF calculations, the active space orbitals are localized on the magnetic atoms. This is accomplished quite readily and the resulting orbitals are localized and orthonormal. In a subsequent calculation, the configurational matrix-elements are obtained.

$$\hat{H} = \sum_i \sum_{P, P' \in i} H_p^{P'} |P'\rangle \langle P| + \sum_{i < j} \sum_{P, P' \in i} \sum_{Q, Q' \in j} H_{pq}^{P'Q'} |P'Q'\rangle \langle PQ| \quad (54)$$

To obtain the one-body matrix element $H_p^{P'}$ as shown in Eq. (54), one simply constructs all neutral configurations $|P\rangle$, and evaluates matrix-elements of the Hamiltonian. By construction only the first two terms in Eq. (3) can contribute. We also evaluate the full two-body Hamiltonian matrix $M_{pq}^{P'Q'}$ over all neutral configurations $|PQ\rangle$. These matrix elements would involve contributions from all elements in Eq. (3) localized on sites i and j . To get the one- and two-body parts of the magnetic Hamiltonian matrix, one defines the explicit two-body part of the Hamiltonian

$$H_{pq}^{P'Q'} = M_{pq}^{P'Q'} - \delta_p^{P'} H_q^{Q'} - H_p^{P'} \delta_q^{Q'}. \quad (55)$$

This procedure is repeated for all pairs of magnetic sites $i < j$ and the matrix elements are then written to a file. This file is read by the program that performs the effective Hamiltonian calculations.

A.2. Definition of partition of $\hat{H}$ into zero-, one- and two-body parts

In principle, the partitioning of interactions into many-body parts is ambiguous, and one can make use of this to define a (small) two-particle fluctuation potential, while a mean-field potential is added to the one-body part of the Hamiltonian. The partitioning is defined in terms of the one-body mean-field density, which satisfies the normalization

$$D(i) = \sum_{I'} D_i^{I'} |I'\rangle \langle I| \quad (56)$$

$$\sum_i D_i^{I'} = 1 \quad (57)$$

The one-body Hamiltonian can be rewritten as

$$\hat{h} = \sum_i \hat{h}_i = E_1 + \sum_i \tilde{h}_i \quad (58)$$

where

$$\tilde{h}_i = \hat{h}_i - E_i \mathbb{1}_i \quad (59)$$

$$E_i = \text{Tr}(h_i D_i) \quad (60)$$

$$E_1 = \sum_i E_i \quad (61)$$

The operator $\tilde{h}_i$ is written in normal order in the sense that

$$\text{Tr}(\tilde{h}_i D_i) = 0 \quad (62)$$

The two-electron interaction can be redefined as

$$\frac{1}{2} \sum_{i \neq j} \hat{V}_{ij} = E_2 + \sum_i \tilde{k}_i + \frac{1}{2} \sum_{a \neq b} \tilde{V}_{ab} \quad (63)$$

where

$$E_2 = \frac{1}{2} \sum_{i \neq j} \sum_{P, P' \in i} \sum_{Q, Q' \in j} V_{pq}^{P'Q'} D_p^P D_q^{Q'} = \frac{1}{2} \sum_{i \neq j} E_{ij} \quad (64)$$

$$\hat{k}_i(j) = \sum_{Q, Q'} V_{pq}^{P'Q'} D_q^{Q'} \quad (65)$$

$$\tilde{k}_i(j) = \hat{k}_i(j) - E_{ij} \mathbb{1}_i \quad (66)$$

Hence

$$\text{Tr}(\tilde{k}_i(j) D_i) = 0 \quad (67)$$

$$\tilde{k}_i = \sum_{j \neq i} \tilde{k}_i(j) \quad (68)$$

$$\tilde{V}_{ij} = \hat{V}_{ij} - \tilde{k}_i(j) \mathbb{1}_j - \mathbb{1}_i \tilde{k}_j(i) + E_{ij} \mathbb{1}_i \mathbb{1}_j \quad (69)$$

Then it follows

$$E_2 + \sum_i \tilde{k}_i + \frac{1}{2} \sum_{i \neq j} \tilde{V}_{ij} = \frac{1}{2} \sum_{i < j} E_{ij} + \sum_i \sum_{j \neq i} (\tilde{k}_i(j) - E_{ij} \mathbb{1}_i \mathbb{1}_j) \quad (70)$$

$$+ \frac{1}{2} \sum_{i \neq j} (\hat{V}_{ij} - \tilde{k}_i(j) \mathbb{1}_j - \hat{k}_j(i) \mathbb{1}_i + E_{ij} \mathbb{1}_i \mathbb{1}_j) \quad (71)$$

$$= \frac{1}{2} \sum_{i \neq j} \hat{V}_{ij} \quad (72)$$

It follows that the full Hamiltonian can be expressed as

$$\hat{H} = (E_1 + E_2) + \sum_i \tilde{f}_i + \frac{1}{2} \sum_{i \neq j} \tilde{V}_{ij} \quad (73)$$

where

$$\tilde{f}_i = \tilde{h}_i + \tilde{k}_i \quad (74)$$

$$\text{Tr}(\tilde{f}_i D_i) = 0 \quad (75)$$

$$\text{Tr}(\tilde{V}_{ij} D_i) = \text{Tr}(D_i \tilde{V}_{ij}) = 0 \quad (76)$$

This operator has all the features that is expressed in a normal ordering with respect to the density matrices D_i . Low-lying energy states are defined as eigenstates of the operator $\tilde{f}_i$.

We will refer to the above procedure as writing the Hamiltonian in normal ordered form with respect to the mean-field density matrices D_p^p . The procedure is analogous to the approach used in conventional Coupled Cluster theory [17], and its generalizations [25–27].

Appendix B. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.cplett.2018.04.017>.

References

- [1] A. Ginsberg, Magnetic exchange in transition metal complexes. 12. Calculation of cluster exchange coupling constants with the X. Alpha-scattered wave method, *J. Am. Chem. Soc.* 102 (1) (1980) 111–117.
- [2] L. Noodleman, Valence bond description of antiferromagnetic coupling in transition metal dimers, *J. Chem. Phys.* 74 (10) (1981) 5737–5743.
- [3] J.P. Malrieu, R. Caballol, C.J. Calzado, C. De Graaf, N. Guihery, Magnetic interactions in molecules and highly correlated materials: physical content, analytical derivation, and rigorous extraction of magnetic Hamiltonians, *Chem. Rev.* 114 (1) (2013) 429–492.
- [4] C. de Graaf, R. Broer, *Magnetic Interactions in Molecules and Solids*, Springer, Switzerland, 2016.
- [5] S.R. White, Density matrix formulation for quantum renormalization groups, *Phys. Rev. Lett.* 69 (19) (1992) 2863.
- [6] U. Schollwöck, The density-matrix renormalization group, *Rev. Mod. Phys.* 77 (1) (2005) 259.
- [7] D. Datta, M. Nooijen, Multireference equation-of-motion coupled cluster theory, *J. Chem. Phys.* 137 (20) (2012) 204107.
- [8] M. Nooijen, L.M. Huntington, O. Demel, D. Datta, L. Kong, K. Shamasundar, V. Lotrich, F. Neese, Communication: multireference equation of motion coupled cluster: a transform and diagonalize approach to electronic structure, *J. Chem. Phys.* 140 (8) (2014) 081102.
- [9] L.M. Huntington, O. Demel, M. Nooijen, Benchmark applications of variations of multireference equation of motion coupled-cluster theory, *J. Chem. Theory Comput.* 12 (1) (2015) 114–132.
- [10] I. Lindgren, J. Morrison, *Atomic Many-body Theory*, vol. 13, Springer Series in Chemical Physics.
- [11] J. Paldus, Coupled cluster theory, in: *Methods in Computational Molecular Physics*, Springer, 1992, pp. 99–194.
- [12] B.O. Roos, P.R. Taylor, P.E.M. Si, A complete active space SCF method (CASSCF) using a density matrix formulated super-cl approach, *Chem. Phys.* 48 (2) (1980) 157–173.
- [13] S. Wu, J. Steffen, P. Hu, M. Nooijen, Multireference equation of motion coupled cluster benchmark study of magnetic model systems, *Comput. Theor. Chem.* 1130 (2018) 160–174.
- [14] G.D. Purvis III, R.J. Bartlett, A full coupled-cluster singles and doubles model: the inclusion of disconnected triples, *J. Chem. Phys.* 76 (4) (1982) 1910–1918.
- [15] Y. Liu, A.D. Dutoi, Excitonic coupled-cluster theory: Part i, general formalism. Available from: <1709.01966>.
- [16] A.D. Dutoi, Y. Liu, Excitonic coupled-cluster theory: Part ii, electronic Hamiltonian. Available from: <1709.01969>.
- [17] R.J. Bartlett, M. Musiał, Coupled-cluster theory in quantum chemistry, *Rev. Mod. Phys.* 79 (1) (2007) 291.
- [18] M. Nooijen, R.J. Bartlett, Similarity transformed equation-of-motion coupled-cluster theory: details, examples, and comparisons, *J. Chem. Phys.* 107 (17) (1997) 6812–6830.
- [19] P.C. Hariharan, J.A. Pople, The influence of polarization functions on molecular orbital hydrogenation energies, *Theor. Chim. Acta* 28 (3) (1973) 213–222.
- [20] M.M. Francl, W.J. Pietro, W.J. Hehre, J.S. Binkley, M.S. Gordon, D.J. DeFrees, J.A. Pople, Self-consistent molecular orbital methods. XXIII. A polarization-type basis set for second-row elements, *J. Chem. Phys.* 77 (7) (1982) 3654–3665.
- [21] E. Olthof, A.v.d. Avoird, P. Wormer, Vibration and rotation of CO in C₆₀ and predicted infrared spectrum, *J. Chem. Phys.* 104 (3) (1996) 832–847.
- [22] A. Krachmalnicoff, R. Bounds, S. Mamone, S. Alom, M. Concistrè, B. Meier, K. Kouřil, M.E. Light, M.R. Johnson, S. Rols, et al., The dipolar endofullerene HF@C₆₀, *Nat. Chem.* 8 (10) (2016) 953–957.
- [23] P.M. Felker, V. Vlček, I. Hietanen, S. FitzGerald, D. Neuhauser, Z. Bačić, Explaining the symmetry breaking observed in the endofullerenes H₂@C₆₀, HF@C₆₀, and H₂O@C₆₀, *Phys. Chem. Chem. Phys.* 19 (46) (2017) 31274–31283.
- [24] J.F. Stanton, J. Gauss, J.D. Watts, W.J. Lauderdale, R.J. Bartlett, The aces ii program system, *Int. J. Quantum Chem.* 44 (S26) (1992) 879–894.
- [25] D. Mukherjee, Normal ordering and a wick-like reduction theorem for fermions with respect to a multi-determinantal reference state, *Chem. Phys. Lett.* 274 (5–6) (1997) 561–566.
- [26] W. Kutzelnigg, D. Mukherjee, Normal order and extended wick theorem for a multiconfiguration reference wave function, *J. Chem. Phys.* 107 (2) (1997) 432–449.
- [27] L. Kong, M. Nooijen, D. Mukherjee, An algebraic proof of generalized wick theorem, *J. Chem. Phys.* 132 (23) (2010) 234107.